Do Now · retrieval
Signed binary and shifts

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

Does changing a by-value int parameter change the caller's int?

Q2

What is a return value?

Q3

Convert 00001010 from unsigned binary to decimal.

Q4

Calculate 00001111 + 00000001 using 8 bits.

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

POSITIVE 13 — +13 written with 8 bits is 00001101.

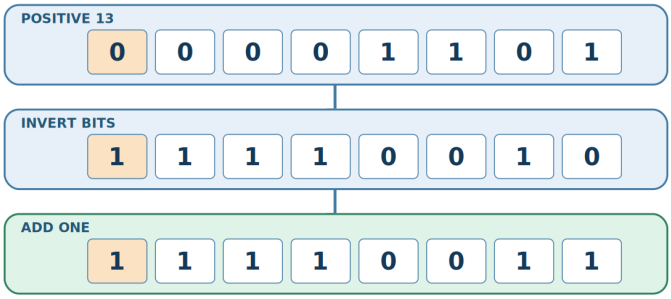
Core idea 2

INVERT BITS — Invert each bit to make 11110010.

Core idea 3

ADD ONE — Add one to get 11110011, the 8-bit two's-complement representation of −13.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

two's complement — Fixed-width signed binary representation.

sign bit — Leftmost bit indicating sign in two's complement.

logical shift — Shift with zero bits entering vacated positions.

arithmetic right shift — Shift right while extending the sign bit.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

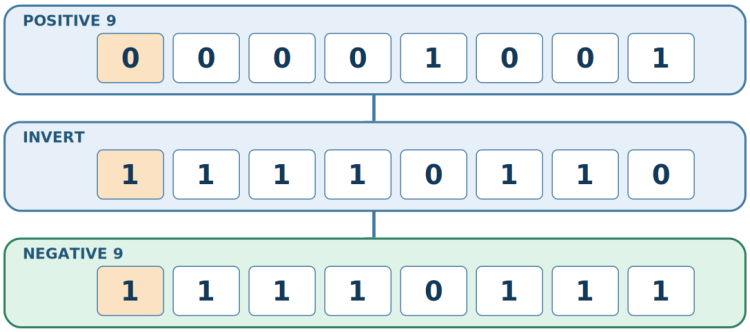
Problem

Find −9 in 8-bit two's complement.

Follow the worked method

1. Predict first — Write +9 in 8 bits and invert all the bits.
2. Model the result — Add 1 to the inverted bits: 11110111 represents −9.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Predict first

+6 = 00000110. Invert to get 11111001.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Write −5 as 8-bit two's complement.

Question 2

Unsigned 8-bit 00110100 is shifted logically right one place. Give the result and decimal value.

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

11110100 represents −12 in two's complement. Arithmetically shift right one place and explain the new left bit.

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

Arithmetic right shift of a negative two's-complement value inserts zero at the left.

Use key vocabulary from memory

1. Explain two's complement in your own words.

2. Explain sign bit in your own words.

3. Explain logical shift in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.